

Optimal Subset Selection in Multi-Agent Image Forensics Ensembles: A Comprehensive Review

Introduction and Problem Formulation

The deployment of multi-agent ensemble systems for image forensics, particularly in resource-constrained edge computing environments, introduces a highly complex combinatorial optimization problem. The core objective is to maximize forensic classification accuracy—distinguishing between pristine and synthetically generated or manipulated media—while strictly minimizing computational overhead, memory footprint, and inference latency. In the specific architectural context under consideration, a four-agent ensemble generates a 43-dimensional feature space. These features are subsequently aggregated and evaluated by a meta-classifier, specifically a Discrepancy-Aware Adaptive Contrastive Learning (DAAC) module, which learns to render final forensic judgments based on the outputs of the constituent agents.

This architecture poses a unique challenge that transcends traditional feature selection. The selection mechanism is inherently modular and agent-level; the elimination of a single agent from the ensemble does not merely remove a single scalar prediction. Instead, it inherently eradicates approximately ten direct feature dimensions, alongside all higher-order pairwise interaction features associated with that specific agent. Consequently, the feature space is structurally inextricably linked to the agent composition.

While a naive enumeration of the solution space—evaluating all $2^4 - 1 = 15$ possible non-empty subsets—is computationally trivial for an ensemble of precisely four agents, this brute-force methodology is fundamentally inadequate from a scientific perspective. First, it lacks the theoretical rigor required to formulate a generalized mathematical contribution to the field of ensemble learning. Second, it fails to scale; as the pool of candidate forensic models expands to encompass dozens or hundreds of specialized agents ($N \gg 4$), the $O(2^N)$ time complexity renders exhaustive search intractable. Most importantly, naive enumeration offers zero explanatory power. It cannot articulate *why* specific agents, or specific combinations thereof, are critical to the meta-classifier's success.

Empirical observations within this specific forensic ensemble reveal a critical anomaly that challenges standard ensemble theory: the *disagreement* between the spatial-frequency agent and the vision-language transformer agent (FatFormer) constitutes the single most

important predictive feature, accounting for an overwhelming 56.5% of the overall feature importance. This high degree of synergy implies that the predictive power of the ensemble is not merely the linear sum of its independent parts. Instead, the value lies deeply embedded within complex, higher-order interactions between heterogeneous models with distinct inductive biases.

This comprehensive report provides an exhaustive, expert-level theoretical analysis of optimal subset selection in multi-agent systems. The analysis is structured across five core research dimensions: the application of cooperative game theory (Shapley valuations) for equitable member valuation, information-theoretic approaches to redundancy and pruning, cost-aware multi-objective optimization for edge deployment, the formulation of agent selection as a combinatorial optimization problem involving submodularity, and the rigorous mathematical modeling of interaction effects. By synthesizing state-of-the-art literature across these domains, this analysis establishes a definitive theoretical framework for scalable, explainable, and hardware-aware agent selection in image forensics.

Research Question 1: Shapley Value for Ensemble Member Valuation

The Shapley value, a foundational solution concept originating from cooperative game theory, provides a mathematically rigorous and axiomatically justified mechanism for fairly distributing the total payoff of a coalition among its individual members based on their marginal contributions. In the context of ensemble machine learning, the "game" is defined as the predictive task, the "players" are the individual models or agents comprising the ensemble, and the "payoff" is the ensemble's overall accuracy, log-likelihood, or reduction in cross-entropy loss.

Data Shapley versus Model Shapley

To appropriately apply game-theoretic valuation to the 4-agent forensics ensemble, a critical theoretical distinction must be established between *Data Shapley* and *Model Shapley*.

Data Shapley frameworks conceptualize individual training data instances as the players in the cooperative game. The primary utility of Data Shapley is to quantify the value of specific data points, enabling practitioners to identify noisy, mislabeled, or exceptionally informative instances to optimize training corpora. Because modern datasets contain millions of instances (M), computing the exact Data Shapley value requires evaluating the model over $O(2^M)$ subsets, which is computationally impossible. Consequently,

extensive literature has been dedicated to developing approximation algorithms. Truncated Monte Carlo (TMC) Shapley leverages random permutation sampling and early stopping to approximate marginal contributions. KNN-Shapley bypasses the need for model retraining entirely by using K-Nearest Neighbors as a surrogate proxy for the underlying deep neural network, computing valuations in near-linear time. Beta-Shapley generalizes the framework further by relaxing the strict efficiency axiom, utilizing a Beta distribution to apply customized, non-uniform weights to subsets of varying cardinalities, thereby reducing noise in the valuation of massive datasets.

Conversely, *Model Shapley* conceptualizes the pre-trained predictive models (the agents) as the players within the coalition. For the forensic ensemble subset selection task, Model Shapley is unequivocally the correct and applicable formulation. The objective is to assign an intrinsic value to the feature generators (the 4 agents), completely independent of the underlying training data instances.

Computational Methods: Exact versus Monte Carlo Approximations for Small Ensembles

When applying Model Shapley to the current architectural configuration, the scale of the player set ($N=4$) fundamentally alters the computational strategy. The exact Shapley value for a model i is defined by the formula:

$$\phi_i = \sum_{S \subseteq N, i \in S} \frac{|S|! (N - |S|)!}{N!} (v(S) - v(S \setminus \{i\}))$$

where N is the set of all agents, S represents a coalition of agents not containing i , and $v(S)$ is the characteristic value function (e.g., the forensic accuracy of the DAAC meta-classifier utilizing only the features produced by subset S). Because $N=4$, calculating the exact Shapley value requires evaluating exactly $2^4 = 16$ subset configurations.

At this scale, employing approximation algorithms such as TMC-Shapley, KNN-Shapley, or Beta-Shapley is mathematically unnecessary and actively detrimental. Monte Carlo approximations introduce variance and estimation errors, which are entirely avoidable when the exact combinatorial space is trivially small. Therefore, for the immediate 4-agent ensemble, exact Model Shapley calculation is the optimal strategy. However, the theoretical value of investigating TMC and KNN approximations lies in the generalizability of the proposed framework. Should the forensic ensemble scale to incorporate $N=50$ heterogeneous agents, exact computation becomes intractable, and the integration of TMC-Shapley or surrogate-based approximations becomes a strictly

necessary component of the subset selection pipeline.

Handling Interaction Effects in Multi-Agent Systems

While the standard Shapley value guarantees a fair distribution of the total ensemble accuracy, it possesses a critical limitation regarding synergistic agents. The Shapley axioms dictate that the value of an interaction (such as the 56.5% importance derived from the disagreement between the Frequency agent and the FatFormer agent) is distributed equally and additively among the participating players. The standard Shapley formulation cannot isolate or explicitly quantify the interaction as a distinct, irreducible entity. If two agents are individually weak but exhibit massive predictive power when paired, their individual Shapley values will increase, but the mathematical output will not explicitly identify the pair as a locked synergistic unit. Addressing this limitation requires advanced game-theoretic indices, which are explored exhaustively under Research Question 5.

Literature Synthesis: Shapley Valuation in Ensembles

Citation	Method Summary	Specific Relevance to Agent Selection
Rozemberczki, B., & Sarkar, R. (2021). The Shapley Value of Classifiers in Ensemble Games. <i>arXiv:2101.02153</i>	Introduces "ensemble games" and the <i>Troupe</i> algorithm, which allocates payoffs based on approximate Shapley values of pre-trained binary classifiers using Monte Carlo sampling.	Directly addresses model-level valuation. Proves that redundant or adversarial models receive low valuations, providing an axiomatic metric for ensemble pruning and subset selection.
Xu, X., Lam, T., Foo, C. S., & Low, B. K. H. (2023). Model Shapley: Equitable Model Valuation with Black-box Access. <i>NeurIPS</i>	Extracts predictive distributions into "Dirichlet abstractions." Defines coalition value via negated Hellinger distance to an oracle. Computes Model	Perfect theoretical fit for the 4-agent ensemble. Dirichlet abstractions elegantly handle heterogeneous agents (Frequency vs. FatFormer) operating in different

Citation	Method Summary	Specific Relevance to Agent Selection
	Shapley via precision-weighted fusion.	feature spaces prior to DAAC fusion.
Rozemberczki, B. et al. (2023). Shapley Values as a Strategy for Ensemble Weights Estimation. <i>Applied Sciences</i>	Utilizes binary cross-entropy for Shapley calculation. Implements a hard pruning strategy where models yielding negative Shapley values are assigned a weight of zero and permanently discarded.	Provides a definitive, mathematically justified threshold for agent exclusion. Any agent in the 4-agent pool yielding a negative Shapley value against the DAAC loss function must be pruned.
Kwon, Y., & Zou, J. (2022). Beta Shapley: a Unified and Noise-reduced Data Valuation Framework. <i>AISTATS</i>	Proposes Beta Shapley by relaxing the efficiency axiom. Utilizes a Beta distribution to assign non-uniform weights to subsets of varying cardinalities to reduce noise.	While designed for Data Shapley, the concept of weighting subset cardinalities is relevant if the DAAC meta-classifier behaves poorly when evaluating excessively small (e.g., 1-agent) subsets.
Jia, R. et al. (2019). Efficient Task-Specific Data Valuation for Nearest Neighbor Algorithms. <i>VLDB</i>	Introduces KNN-Shapley, bypassing model retraining by using K-Nearest Neighbors as a surrogate to compute exact Shapley valuations in near-linear time.	Highly relevant for scaling the framework to \$N \gg 4\$. If the agent pool expands, evaluating DAAC 10,000 times is impossible; a KNN surrogate over the 43-D feature space offers a scalable alternative.

The synthesis of these foundational papers establishes that Model Shapley, explicitly calculated via the DAAC meta-classifier's loss function, provides the ultimate, equitable baseline for determining the individual marginal utility of the image forensics agents.

Research Question 2: Information-Theoretic Approaches to Ensemble Pruning

While game-theoretic valuations provide an axiomatic distribution of overall model performance, information theory provides a rigorous mathematical vocabulary for quantifying the precise amount of uncertainty reduced by the inclusion of a specific agent. By analyzing the flow of information between the 43-dimensional agent outputs and the ground-truth forensic labels, one can establish hard theoretical bounds on an agent's utility, isolating the concepts of predictive relevance and representational redundancy.

Conditional Mutual Information for Measuring Agent Contribution

In standard feature selection, the fundamental objective is to maximize relevance to the target while minimizing redundancy among the selected features. Shannon's Mutual Information, denoted as $I(X; Y)$, measures the total reduction in uncertainty regarding the target label Y given the observation of an agent's output X . However, evaluating agents solely on their individual mutual information is highly deceptive. An agent with a remarkably high $I(X; Y)$ may be entirely redundant if another agent Z , which captures the exact same variance in the target variable, is already present within the ensemble.

To accurately measure an agent's unique contribution to the ensemble, one must utilize Conditional Mutual Information (CMI). CMI, defined mathematically as $I(X; Y | Z) = H(Y | Z) - H(Y | X, Z)$, measures the *novel* information that agent X provides about the ground-truth forensic label Y , given that the outputs of the remaining ensemble Z are already known and conditioned upon.

In the specific context of the 4-agent DAAC meta-classifier, evaluating $I(X_{\text{agent}}; Y | X_{\text{ensemble} \setminus \text{agent}})$ perfectly captures the unique, irredundant forensic artifacts that a specific agent detects. For example, if the Frequency agent captures high-frequency diffusion upsampling noise that the spatial agents entirely miss, its CMI will be exceptionally high. Conversely, if two spatial CNN agents extract identical edge-level artifacts, the CMI of the second CNN, conditioned on the first, will approach zero, indicating that the entire 10-dimensional feature block of the second CNN can be safely pruned from the DAAC input space without any loss of forensic classification capacity.

To stabilize these measurements and ensure that the selection accounts for

complementary interactions, Joint Mutual Information (JMI) can be employed. JMI evaluates the information between the target Y and a joint random variable formed by pairing a candidate agent's features with previously selected features in the ensemble, directly focusing on increasing complementary information.

Redundancy Analysis via Centered Kernel Alignment (CKA)

While CMI evaluates redundancy relative to the final ground-truth label, representation similarity metrics allow for the analysis of redundancy directly between the internal mechanisms of the agents themselves, entirely independent of the DAAC meta-classifier.

Centered Kernel Alignment (CKA) is a state-of-the-art metric that quantifies the alignment between neural representations using centered Gram matrices and the Hilbert-Schmidt Independence Criterion. Unlike canonical correlation analysis (CCA), CKA achieves strict invariance to orthogonal transformations and isotropic scaling, making it highly effective for comparing the covariance structures of wildly different neural architectures.

By extracting the activation tensors from the penultimate layers of the four agents and computing the pairwise CKA scores, practitioners can map the representational topology of the ensemble. If a standard CNN agent and a novel Transformer agent exhibit a CKA score approaching 1.0, it mathematically proves that their internal feature manifolds are functionally identical, despite their differing architectural paradigms. In such a scenario, the agent with the higher computational latency can be immediately pruned. Conversely, a low CKA score between the Frequency agent and the FatFormer agent confirms that they are operating on orthogonal manifolds, predicting the high interaction effect observed in the DAAC feature importance analysis.

Literature Synthesis: Information Theory in Pruning

Citation	Method Summary	Specific Relevance to Agent Selection
Brown, G., Pocock, A., Zhao, M. J., & Luján, M. (2012). A unifying framework for information theoretic	Unifies two decades of heuristic filter criteria under the conditional likelihood of training labels. Establishes the JMI	Because dropping one agent removes ~10 features simultaneously, this framework adapts perfectly to block-feature

Citation	Method Summary	Specific Relevance to Agent Selection
feature selection. <i>JMLR</i>	criterion as the optimal formulation for balancing relevance and redundancy.	selection, justifying the retention of agents that maximize JMI with the DAAC meta-classifier.
Steinke, T., & Zakynthinou, L. (2020). Reasoning About Generalization via Conditional Mutual Information. <i>COLT</i>	Provides an information-theoretic framework using CMI to quantify generalization bounds for machine learning algorithms, superseding traditional uniform convergence bounds.	Ensures the selected agent subset will generalize to unseen forensic perturbations. Agents exhibiting low CMI on validation sets are mathematically guaranteed to degrade generalization, mandating their exclusion.
Kornblith, S., Norouzi, M., Lee, H., & Hinton, G. (2019). Similarity of neural network representations revisited. <i>ICML</i>	Establishes CKA as a robust variance-weighted metric capable of reliably identifying corresponding layers and redundant representational pathways across disparate network architectures.	Allows for pre-training pruning. Computing pairwise CKA on the internal representations of the 4 agents instantly identifies structural redundancy before the DAAC meta-classifier is even initialized.
Gadgil, S., Covert, I., & Lee, S.-I. (2023). Estimating Conditional Mutual Information for Dynamic Feature Selection.	Prioritizes features based on mutual information with the response variable, proposing a novel approach that estimates CMI in a	Provides a modern, computationally tractable methodology for estimating the exact CMI of the 43-dimensional agent features in the

Citation	Method Summary	Specific Relevance to Agent Selection
<i>arXiv:2306.03301</i>	discriminative rather than generative fashion.	highly non-linear forensic classification space.

Integrating these information-theoretic approaches transitions the ensemble pruning methodology from empirical guesswork to a formal mathematical discipline, ensuring that only agents providing statistically independent, high-entropy forensic signals are retained for edge deployment.

Research Question 3: Cost-Aware Model/Module Selection

The deployment of multi-agent forensic systems in edge computing environments—such as autonomous drones, mobile devices, or localized IoT sensors—imposes uncompromising hardware constraints. Parameters such as inference latency, memory footprint, and energy consumption are as critical as forensic accuracy. Optimizing the DAAC meta-classifier solely for accuracy will invariably result in the selection of the full, maximum-capacity $N=4$ ensemble. Therefore, agent selection must be formally framed as a Multi-Objective Optimization (MOO) problem.

Pareto-Optimal Ensemble Selection and Evolutionary Algorithms

In multi-objective optimization, a solution is considered *Pareto optimal* if no other valid subset of agents can achieve higher forensic accuracy without simultaneously incurring a greater computational cost (e.g., latency or memory). The complete set of these non-dominated solutions forms the Pareto front. The objective of cost-aware selection is to present the systems engineer with this precise mathematical boundary, allowing them to select an ensemble that maximizes accuracy while fitting within a strict, pre-defined operational budget.

To navigate this objective space, evolutionary algorithms such as NSGA-II (Non-dominated Sorting Genetic Algorithm II) are universally employed. NSGA-II maintains and evolves a population of candidate ensembles across generations. It evaluates each candidate against both accuracy and cost, utilizing a fast non-dominated sorting mechanism to rank the population into Pareto fronts. Crucially, it employs a crowding distance metric to preserve diversity, ensuring that the final output provides a wide spread of options across the accuracy-cost spectrum. While deploying an evolutionary

algorithm to search the 15 possible subsets of a 4-agent ensemble is computationally excessive, establishing the NSGA-II framework is vital for generalizing the methodology to future iterations where the agent pool is vastly expanded.

Bayesian Multi-Objective Optimization

An alternative to evolutionary search is Multi-Objective Bayesian Optimization (MOBO). Evaluating the accuracy of a specific agent subset requires executing the full inference pipeline and querying the DAAC meta-classifier, which is a computationally expensive black-box operation. MOBO addresses this by fitting a probabilistic surrogate model (typically a Gaussian Process) to predict the accuracy and cost of untested subsets based on prior evaluations.

In scenarios where objective functions exhibit radically different evaluation costs—for instance, calculating an agent's parameter size is instantaneous, whereas computing its forensic accuracy requires extensive forward passes—standard MOBO is inefficient. Advanced formulations, such as Flexible Multi-Objective Bayesian Optimization (FlexiBO), resolve this by dynamically weighting the acquisition function (e.g., Expected Hypervolume Improvement) against the measurement cost of the objective. This ensures that the algorithm extracts the maximum possible information regarding the Pareto front per unit of computational expenditure.

Differentiable Agent Selection via Neural Architecture Search (NAS)

Perhaps the most mathematically elegant approach to cost-aware module selection is to embed the selection process directly into the training loop of the meta-classifier, effectively treating ensemble pruning as a Neural Architecture Search (NAS) problem.

The fundamental challenge in NAS is that architectural decisions (such as choosing whether to include or exclude a specific agent's 10-dimensional feature block) are discrete, categorical, and therefore non-differentiable. This breaks standard backpropagation. To circumvent this, the Gumbel-Softmax reparameterization trick is employed. Gumbel-Softmax relaxes the discrete categorical sampling into a continuous, temperature-controlled softmax distribution, enabling low-variance gradient estimation.

By applying a Gumbel-Softmax mask to the 43-dimensional input vector of the DAAC classifier, the network can smoothly interpolate between including and excluding specific agents. By appending a hardware-penalty term (e.g., cumulative FLOPs) directly to the binary cross-entropy loss function, the network becomes strictly cost-aware. As the temperature parameter τ is annealed toward zero during training, the soft

probability distributions harden into discrete binary choices. The meta-classifier thus autonomously learns to drop the most computationally expensive, least informative agents, solving the combinatorial subset selection problem end-to-end via pure gradient descent.

Literature Synthesis: Cost-Aware Optimization

Citation	Method Summary	Specific Relevance to Agent Selection
Maier, J., Möller, F., & Purucker, L. (2024). Hardware Aware Ensemble Selection for Balancing Predictive Accuracy and Cost. <i>AutoML</i>	Extends Quality Diversity Optimization (QDO-ES) by replacing standard diversity metrics with hardware cost metrics (inference time, size) to generate an accuracy-efficiency Pareto front.	Directly addresses the edge-deployment constraint. Proves that tracking subsets along a Pareto front allows practitioners to mathematically justify discarding an agent if its marginal accuracy gain demands unacceptable latency.
Iqbal, M. S., Su, J., Kotthoff, L., & Jamshidi, P. (2020). Flexible Multi-Objective Bayesian Optimization. <i>JAI/R</i>	Formulates an acquisition function that weights the improvement of the Pareto hypervolume by the actual measurement cost of each objective, maximizing information gain per unit cost.	** By merging hypervolume knowledge gradients (information theory) with decoupled evaluation costs, this framework guarantees the most efficient traversal of the ensemble subset space.**
Chang, J., Zhang, X., et al. (2019).	Replaces non-differentiable argmax	Enables end-to-end selection. By masking the

Citation	Method Summary	Specific Relevance to Agent Selection
Differentiable Architecture Search with Ensemble Gumbel-Softmax. <i>arXiv:1905.01786</i>	operations with a continuous Gumbel-Softmax estimator, allowing structural architectural decisions to be optimized concurrently with model weights.	DAAC inputs with Gumbel-Softmax variables and penalizing latency in the loss function, the network autonomously prunes costly agents during backpropagation.
Deb, K., Pratap, A., Agarwal, S., & Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. <i>IEEE Trans. Evol. Comput.</i>	Proposes a fast non-dominated sorting approach with elitism and a crowding distance metric to maintain diversity along the Pareto front.	Serves as the foundational baseline for establishing the Pareto front of the forensic ensemble subsets, vital for generalizing the framework to future $N > 4$ configurations.

Research Question 4: Agent Selection as a Combinatorial Optimization Problem

Selecting the optimal subset of agents subject to a strict computational budget is fundamentally a discrete combinatorial optimization problem. When framed directly, it is a variation of the 0-1 Knapsack problem: the algorithm must select a subset of items (agents) to maximize total value (forensic accuracy) such that the total weight (inference latency or memory) does not exceed a predefined budget constraint. The 0-1 Knapsack problem is inherently NP-hard, necessitating approximation strategies as the agent pool expands.

Submodular Function Maximization

A highly powerful mathematical framework for approximating combinatorial subset selection relies on the concept of *submodularity*. A set function $f: 2^V \rightarrow \mathbb{R}$ is defined as submodular if it exhibits the property of diminishing marginal returns. Formally, for every $A \subseteq B \subseteq V$ and for every element $x \in V$

W ⊆ B, the function satisfies:

$$f(A \cup W) - f(A) \geq f(B \cup W) - f(B)$$

In the context of the forensics ensemble, submodularity implies that adding a new, highly accurate agent to a small, weak ensemble will yield a massive increase in overall accuracy. However, adding that exact same agent to a large, highly capable ensemble will yield a significantly smaller, diminishing marginal improvement.

If the objective function (e.g., the DAAC meta-classifier's accuracy or the Joint Mutual Information) is demonstrably non-negative, monotone, and submodular, a foundational theorem by Nemhauser et al. guarantees that a simple, computationally inexpensive greedy algorithm will perform exceptionally well. By iteratively adding the single agent that provides the maximum marginal benefit relative to its cost, the greedy algorithm is mathematically proven to achieve a solution that is at least $(1 - 1/e)$ —approximately 63.2%—of the absolute optimal subset.

The Submodularity Ratio and Synergistic Violations

A critical theoretical question arises: Is the forensic ensemble accuracy function actually submodular?

The empirical evidence derived from the 4-agent system strongly suggests that it is *not* strictly submodular. The observation that the disagreement between the Frequency agent and the FatFormer agent constitutes 56.5% of the total feature importance implies the existence of a massive synergistic interaction. In cooperative game theory and combinatorial optimization, synergy is the antithesis of diminishing returns; it creates *supermodularity*. Evaluating the Frequency agent alone yields low accuracy, evaluating FatFormer alone yields moderate accuracy, but combining them yields an explosive, non-linear jump in performance. This violates the submodular inequality.

To address functions that are only weakly submodular due to local supermodular synergies, researchers utilize the *submodularity ratio* (γ). This ratio quantifies exactly how "close" a given set function is to being strictly submodular. If a function possesses a submodularity ratio γ , the greedy approximation guarantee degrades gracefully from $(1 - 1/e)$ to $(1 - e^{-\gamma})$. By analyzing the eigenvalues of the feature covariance matrix, practitioners can bound this ratio. Establishing the submodularity ratio of the DAAC accuracy function provides the necessary theoretical proof to justify using greedy selection heuristics despite the presence of complex, interacting agents.

Literature Synthesis: Combinatorial Optimization

Citation	Method Summary	Specific Relevance to Agent Selection
Das, A., & Kempe, D. (2011). Submodular meets Spectral: Greedy Algorithms for Subset Selection. <i>ICML</i>	Introduces the submodularity ratio (ω_{gamma}) to measure deviation from strict submodularity. Proves that greedy algorithms provide a $(1 - e^{-\omega_{\text{gamma}}})$ approximation, bounding ω_{gamma} via sparse eigenvalues.	Crucial for theoretical rigor. Because the Frequency-FatFormer synergy violates strict submodularity, calculating ω_{gamma} formally bounds the worst-case performance of any greedy subset selection deployed on the ensemble.
Buyukates, B., et al. (2025). Don't Always Pick the Highest-Performing Model: An Information Theoretic View of LLM Ensemble Selection. <i>arXiv:2602.08003</i>	Frames budgeted ensemble selection as maximizing mutual information. Models correlated model errors using Gaussian-copulas to explain performance saturation, isolating a submodular "difficulty-aware" component.	Explains why naive selection fails. Two weak agents (Frequency/FatFormer) with uncorrelated error distributions (modeled via copulas) yield higher subset mutual information than two highly accurate, highly correlated CNNs.
Bach, F. (2013). Learning with Submodular Functions: A Convex Optimization Perspective.	Presents the theory of submodular functions from a convex analysis perspective. Explores the Lovász extension of submodular functions to formulate regularizers for	Provides the mathematical foundation for converting the discrete combinatorial subset search into a continuous, convex optimization problem via

Citation	Method Summary	Specific Relevance to Agent Selection
<i>Foundations and Trends in Machine Learning</i>	supervised learning.	the Lovász extension.
Wei, K., et al. (2015). Submodularity in data subset selection and active learning. <i>ICML</i>	Makes a formal connection between submodular functions and accuracy functions for data/feature subset selection, establishing log-likelihood score functions as approximately submodular.	Supports the hypothesis that the overall objective function of the DAAC meta-classifier, despite local synergies, operates within an approximately submodular framework globally.

Research Question 5: Interaction Effects in Ensemble Features

The optimization of the multi-agent system hinges on a singular, dominant empirical finding: the *disagreement* between the Frequency agent and the FatFormer agent is the most critical feature utilized by the DAAC meta-classifier, commanding 56.5% of the total feature importance. This indicates that the core forensic capability of the system resides not within the individual predictive power of any single agent, but specifically within the conceptual delta between spatial-semantic interpretation (driven by the Vision-Language transformer) and low-level spectral artifact detection (driven by the Frequency analysis module). Agent interactions, therefore, matter vastly more than individual agent performance.

Limitations of Standard Attribution and the Need for Interaction Indices

As established in RQ1, the standard Shapley value is an elegant tool for assigning linear, additive credit to individual players. However, it is fundamentally incapable of isolating higher-order interactions. The axioms that govern the standard Shapley value forcefully distribute the synergistic value of the Frequency-FatFormer interaction equally among the two agents. While this accurately reflects their individual importance, it masks the structural dependency between them; it fails to alert the pruning algorithm that dropping *either* agent will catastrophically collapse the meta-classifier's primary decision

boundary.

To map and preserve this topology, the framework must utilize advanced interaction indices derived from cooperative game theory.

1. Shapley Interaction Index (SII): An early extension of the standard Shapley value, the SII allocates credit not merely to individual players, but to pairs (or larger subsets) of players, capturing local interaction effects. While foundational, the SII suffers from a severe mathematical flaw: it fails to satisfy the efficiency axiom for higher-order interactions. Consequently, the sum of all main effects and interaction effects does not perfectly equal the total predictive output of the model, leading to inaccurate attributions that can mislead subset selection.

2. Shapley-Taylor Interaction Index (STII): To resolve the mathematical inconsistencies of the SII, the Shapley-Taylor Interaction Index was formulated. The STII constructs the attribution problem by drawing an exact, elegant analogy to the truncated Taylor Series expansion of a function. It decomposes the value function's multilinear extension into discrete derivatives evaluated at different coalition points. Crucially, the STII introduces a novel *interaction distribution axiom* alongside the standard Shapley axioms (Linearity, Dummy, Symmetry, and Efficiency). This allows the STII to isolate pure main effects (self-interactions) from genuine, higher-order interactions of arbitrary size k . By configuring the STII to $k=2$, the algorithm will definitively isolate and quantify the massive 56.5% synergistic weight of the {Frequency, FatFormer} pair.

3. Faithful Shapley Interaction Index (Faith-Shap): A contemporary alternative to STII, Faith-Shap addresses the interaction problem without requiring the introduction of novel, potentially conflicting axioms. Instead, it frames Shapley values strictly as the coefficients of the most faithful polynomial approximation to the pseudo-Boolean coalition game value function. By extending this from a linear to a higher-order polynomial approximation, Faith-Shap yields an interaction index through the resolution of a highly efficient, weighted linear regression problem.

Preserving Critical Interactions During Pruning

Once the STII or Faith-Shap index has mathematically isolated the massive interaction weight of the {Frequency, FatFormer} subset, the cost-aware pruning algorithms (such as the Gumbel-Softmax NAS or the NSGA-II MOO) must be constrained to preserve it. This can be achieved by algorithmically fusing the two agents into a single "macro-agent" or "super-node" within the subset selection graph. By forcing the combinatorial optimizer to

treat the highly synergistic pair as an indivisible binary unit, the system guarantees that the foundational decision boundary of the DAAC meta-classifier is protected, even under the strictest edge-deployment latency budgets.

Literature Synthesis: Interaction Effects

Citation	Method Summary	Specific Relevance to Agent Selection
Sundararajan, M., Dhamdhere, K., & Agarwal, A. (2020). The Shapley Taylor Interaction Index. <i>ICML</i> , PMLR 119	Proposes STII, attributing predictions to feature subset interactions up to size k . Axiomatized using an explicit interaction distribution axiom, resolving the inefficiency of the traditional SII.	Provides the precise mathematical tool to prove <i>why</i> the Frequency and FatFormer agents must be retained. Computing STII at $k=2$ isolates and explains the 56.5% importance of their disagreement.
Tsai, C. P., Yeh, C. K., & Ravikumar, P. (2023). Faith-Shap: The Faithful Shapley Interaction Index. <i>JMLR</i> , 24(94), 1-42	Defines interaction indices as coefficients of the most faithful higher-order polynomial approximation to the coalition game, calculated efficiently via weighted linear regression.	Offers a computationally superior alternative to STII for evaluating the 4-agent ensemble. The regression coefficients directly quantify the interaction strength of the 43-D pairwise features.
Liu, Z., et al. (2024). Forgery-aware Adaptive Transformer for Generalizable Synthetic Image	Introduces FatFormer, which adapts frozen Vision-Language models (CLIP) with forgery-aware adapters to integrate both spatial-semantic	Explains the forensic mechanism driving the interaction. FatFormer captures high-level semantic anomalies, while the dedicated Frequency agent

Citation	Method Summary	Specific Relevance to Agent Selection
Detection. <i>CVPR</i>	and high-frequency forensic traces.	captures low-level spectral noise. Their resulting disagreement forms an ultimate, robust meta-feature.
Grabisch, M., & Roubens, M. (1999). An axiomatic approach to the Banzhaf interaction index. <i>International Journal of Game Theory</i>	Explores the Banzhaf interaction index for measuring the interaction degree inside coalitions, defining it as a solution to a standard least squares approximation for pseudo-Boolean functions.	While superseded by STII and Faith-Shap in modern interpretability, this establishes the foundational game-theoretic proofs for attributing value to coalitions rather than individual players.

Conclusion

The objective of selecting the optimal subset of agents within a resource-constrained, 4-agent image forensics ensemble cannot be satisfied by heuristic subset enumeration. To establish a rigorous, scalable, and explainable methodology, the problem must be deconstructed through a synthesis of game theory, information theory, and continuous optimization.

The theoretical framework must first reject Data Shapley in favor of **Model Shapley**, utilizing precise coalition evaluations to establish the baseline marginal utility of each agent. However, because standard Shapley masks synergy, the **Shapley-Taylor Interaction Index (STII)** or **Faith-Shap** must be applied to mathematically isolate the massive 56.5% interaction effect generated by the disagreement between the Frequency and FatFormer agents. Prior to training the DAAC meta-classifier, structural redundancy among the agents should be mapped and eliminated using **Centered Kernel Alignment (CKA)** and **Conditional Mutual Information (CMI)**.

Finally, navigating the accuracy-cost trade-off must be framed as a Multi-Objective

Optimization problem. By calculating the **submodularity ratio** (γ) of the meta-classifier's accuracy function, practitioners can mathematically bound the performance of greedy knapsack selection. To achieve true end-to-end optimization, **Differentiable Architecture Search via Ensemble Gumbel-Softmax**, augmented by **Cost-Aware Bayesian Optimization (FlexiBO)**, should be deployed. This allows the meta-classifier to autonomously penalize latency and prune costly agents during gradient descent, ensuring the final ensemble architecture is unequivocally Pareto-optimal, deeply explainable, and seamlessly deployable at the edge.